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Methane diffusion path in HKUST-1 metal-organic framework revealed by atomistic simulations

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A copper-based metal-organic framework (MOF) HKUST-1 exhibits high methane adsorption capacity, primarily due to strong Coulomb interactions near open metal sites (OMSs) and van der Waals interactions within ligand-enclosed cavities. While the adsorption sites have been widely studied, the diffusion pathways through which methane reaches these sites remain unclear, particularly regarding the influence of OMSs. In this study, molecular dynamics (MD) simulations were conducted to investigate methane diffusion in HKUST-1 with and without OMSs. First-principles calculations and methane uptake experiments were also performed to support and validate the simulation results. We found that methane was trapped in ligand-enclosed sites due to steric hindrance, while regions near OMSs serve as diffusion hubs. Moreover, in the absence of OMSs, methane backflow was observed at the surface of HKUST-1 due to the presence of stable adsorption sites on surface planes. When OMSs are present, these stable sites shift toward the OMS regions via Coulomb interactions, reducing surface backflow and enhancing methane uptake. The experimentally measured increase in methane adsorption for HKUST-1 with OMSs supports the simulation results. This study can guide the future design of MOFs with enhanced gas adsorption capacity.

Keywords HKUST-1, Methane diffusion, MD simulation, First-principles calculation, Experiment

Methane, a greenhouse gas with a global warming potential 28 times higher than that of carbon dioxide[1], must be removed to mitigate climate change. Removing methane through capture can contribute to the mitigation, and the captured methane can also be repurposed as fuel, such as biogas. Adsorbents enable methane capture; however, the process is challenging owing to the high symmetry and low polarity of methane. Recent studies have shown that metal-organic frameworks (MOFs) exhibit superior methane adsorption capacities compared with conventional adsorbents such as zeolites and activated carbons[2]. MOFs are nanoporous materials comprising metal ions and organic ligands, which form structures with tunable pore sizes and geometries[3]. These structural characteristics, along with their high specific surface area resulting from the framework structures, make MOFs a promising alternative for gas storage via adsorption[4,5].

Among MOFs, $Cu_3(BTC)_2$ (BTC = 1,3,5-benzene tricarboxylate), commonly known as HKUST-1, exhibits considerably high methane adsorption capacity[5,6]. This high adsorption capacity can be attributed to the presence of open metal sites (OMSs), which are created during activation in which unstable ligands or atoms bonded to metals are removed by solvents and heat and vacuum treatment[7,8]. These sites are unsaturated and charge-dense, so are thought to bind gas molecules more strongly than other adsorption sites[9]. HKUST-1 comprises paddle-wheel coordination units, each comprising two Cu^{2+} ions and four BTC linkers. Upon activation, OMSs are formed at the Cu centers in HKUST-1 (referred to as activated HKUST-1) along the axial direction of the paddle-wheel units.

To determine whether these OMSs contribute to methane adsorption, Getzschmann et al. performed an in situ neutron diffraction analysis and identified that methane was frequently observed near OMSs, but it was also observed in abundance in sites far from OMSs[7]. The causes of this methane distribution were investigated by Getzschmann et al.[7] and Wu et al.[10] using grand-canonical Monte Carlo (GCMC) simulation with the universal force field (UFF)[11]. Both simulations succeeded to predict the methane distribution in sites far from OMSs, but failed to predict the high methane concentration near OMSs[7,10]. This was due to the limitation of UFF that it

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cannot precisely capture the interactions between metal atoms and fluid molecules at distances much shorter than their combined hard-sphere diameter[7,10]. To address the limitation of UFF, Chen et al. incorporated a potential energy surface derived from hybrid density functional theory (DFT) and ab initio calculations into their GCMC simulations[8], and Koh et al. developed a force field that accurately describes the methane-OMS interaction through quantum-mechanical calculations[12]. Hence, both simulations[8,12] reproduced the high methane concentration around OMSs similar to the experimental results reported by Getzschmann et al.[7]. Through these studies[7,8,10], it was confirmed that methane is mainly adsorbed at OMSs with strong Coulomb attraction and at van der Waals potential pocket sites where dispersive interactions are amplified by contact with multiple surfaces. These van der Waals-amplified sites are found in small cages and their windows, while large cages show little methane uptake.

The findings from previous studies provide valuable insights for improving the methane storage capacity of MOFs, but they are limited by the fact that the analyses were conducted using GCMC simulations that explore equilibrium states. While these studies identified the preferential adsorption sites depending on methane loading, they could not capture the dynamic pathways by which methane molecules migrate and occupy these sites. This is important considering that the causes of adsorption near and far from OMS are different, suggesting that methane diffusion pathways may also vary depending on the site type. Furthermore, in the case of inactivated HKUST-1 before the activation process, there is no OMS at the Cu center. In this case, since there is no Coulomb attraction due to the absence of OMS, it is necessary to confirm whether the methane concentration near the Cu center decreases and whether this causes a change in the overall diffusion behavior of methane in inactivated HKUST-1.

One approach to examine methane diffusion behavior in HKUST-1 is through molecular dynamics (MD) simulations, which track the movement of methane molecules over time. In this study, we used MD simulations to investigate diffusion behavior, including methane diffusion coefficients and the diffusion pathways of methane molecules from the surface to adsorption sites in HKUST-1. To assess the influence of OMSs on methane diffusion, these methods were applied to both activated HKUST-1 (containing OMSs) and inactivated HKUST-1 (lacking OMSs), and the results were compared. Additionally, first-principles calculations were performed to determine the methane adsorption energies and equilibrium distances in both HKUST-1, providing insights into the diffusion mechanism. Finally, to validate the simulation results, methane uptake in activated and inactivated HKUST-1 was experimentally measured under identical conditions.

Results and discussion

Diffusion coefficient of methane in HKUST-1

To examine methane diffusion behavior in inactivated and activated HKUST-1, we first calculated the diffusion coefficient, a fundamental property of diffusion, through MD simulations. In the three-dimensional MD simulations, the normal diffusion coefficient D[13] is defined as follows:

$$D = \lim_{t \to \infty} \frac{1}{6t} \left\langle \left[\overrightarrow{\mathbf{r}}(t) - \overrightarrow{\mathbf{r}}(0) \right]^2 \right\rangle \tag{1}$$

where $\overrightarrow{\mathbf{r}}(t)$ and $\left\langle \left[\overrightarrow{\mathbf{r}}(t) - \overrightarrow{\mathbf{r}}(0) \right]^2 \right\rangle$ are the position of the methane molecule and MSD at time t, respectively. To determine the diffusion coefficients, we calculated the MSD of methane molecules in inactivated and activated HKUST-1 at 300 K for 20 ns. The slope of the curve of MSD versus time for activated HKUST-1 was approximately twice that for inactivated HKUST-1 (Fig. 1a), indicating faster methane diffusion in activated HKUST-1. To obtain normal diffusion coefficients from the slopes using Eq. (1), the MSD must exhibit linear behavior over time[14], which was verified by applying logarithmic functions to the MSD and time and identifying whether the slope of the curve of log(MSD) versus log(time) approached one. As shown in Fig. 1b, inactivated and activated HKUST-1 exhibited sublinear slopes during the early diffusion stage, known as anomalous diffusion[14], which arose from the pore morphology of HKUST-1. However, as the diffusion progressed, the slopes approached one (for inactivated and activated HKUST-1). This diffusion behavior was observed in a previous study on methane diffusion in coal[15]. In Fig. 1b, the transition from anomalous diffusion to normal diffusion was observed at log(time) ≈ 2.5 (time ≈ 316 ps). Therefore, the diffusion coefficients were determined from the slope of the MSD versus time using only the data within the normal diffusion regime after 316 ps (Fig. 1a). The resulting normal diffusion coefficients for inactivated and activated HKUST-1 were 46.00×10^{-10} and 99.92×10^{-10} m^2/s, respectively. Notably, these values were significantly lower than those reported by Babaei et al.[16] for methane diffusion in activated HKUST-1 ($797–1107 \times 10^{-10}$ m^2/s). This discrepancy likely arises from the difference in the number of methane molecules (135 molecules in this study vs. 740–2220 molecules per simulation box in the study by Babaei et al.) and the diffusion coefficient calculation methods (MSD analyses using 3 × 3 × 3 unit cell samples with periodic boundary conditions in all directions in this study vs. transient mass curve analyses using 4–12 × 1 × 1 unit cell samples with surfaces on (100) and (_100) planes in the study by Babaei et al.). However, a study with simulation conditions similar to this study reported a methane diffusion coefficient of 0.401×10^{-10} m^2/s in ZIF-8[17]. Because ZIF-8 exhibits a lower methane adsorption capacity[18], the present calculations are expected to align with the experimental trends.

Diffusion path of methane in HKUST-1

The MSD analysis in Fig. 1 illustrates methane diffusion within the interior of HKUST-1. However, during the actual diffusion process, methane initially existed in the external space before entering HKUST-1. To simulate a more realistic diffusion process, we performed MD simulations to examine the entry of methane into the inactivated and activated HKUST-1 samples with surfaces exposed on the (_100) and (100) planes. These surface

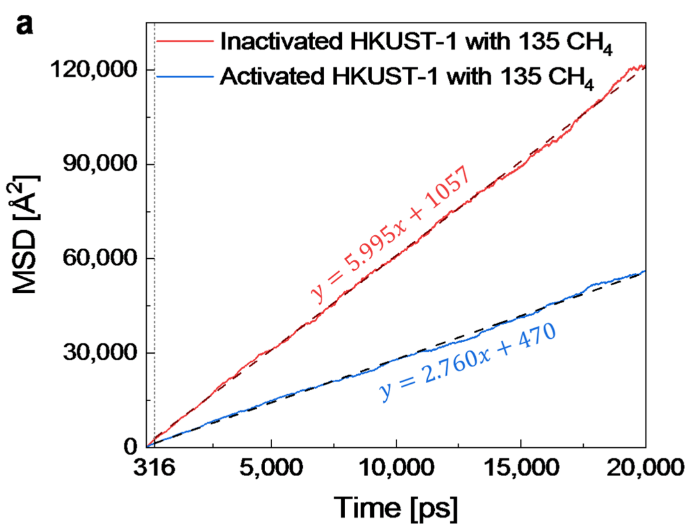

Fig. 1. Plots of (**a**) mean square displacement (MSD) vs. time and (**b**) log(MSD) vs. log(time) for inactivated and activated HKUST-1 at 300 K. The vertical dotted line marks the transition from anomalous to normal diffusion. (**a** and **b**) The dashed line corresponds to the slope in the normal diffusion regime, while (**b**) the dotted line corresponds to the slope in the anomalous diffusion regime.

planes were chosen based on the crystal structure of HKUST-1, which belongs to the *Fm_3m* space group, corresponding to the NaCl-type structure with stable {100} surface planes. We introduced an empty space adjacent to the samples, added methane molecules, and performed an NVT simulation at 300 K for 200 ps, during which the methane adsorption behavior appeared to reach near-convergence. Animations illustrating methane diffusion in inactivated and activated HKUST-1 with 100 methane molecules at 300 K are shown in Fig. S1. To assess the overall methane behavior, we counted the number of methane molecules entering HKUST-1 over time. As shown in Fig. 2a, the number of methane molecules entering HKUST-1 increased rapidly before slowing. In all cases, methane diffusion was faster in activated HKUST-1 than in inactivated HKUST-1.

To gain deeper insight into the methane diffusion behavior, we counted the number of methane molecules along the thickness direction of HKUST-1 over time. As shown in Fig. 2b,c and Fig. S2, the methane molecules diffused from the edges to the center of HKUST-1. The diffusion rate was higher in activated HKUST-1 than in inactivated HKUST-1, which is consistent with the diffusion coefficient calculations shown in Fig. 1. However, the diffusion was non-uniform because certain regions contained more methane molecules than others. This uneven distribution was particularly pronounced in activated HKUST-1. We hypothesize that the observed non-uniform diffusion is induced by the pore morphology of HKUST-1, similar to the anomalous diffusion behavior seen during the early stages of diffusion coefficient calculation in Fig. 1b. Furthermore, the pronounced uneven distribution of methane in activated HKUST-1 is believed to result from Coulomb interactions between methane molecules and OMSs.

To test these hypotheses, we analyzed the diffusion path of each methane molecule in inactivated and activated HKUST-1. Particularly, we identified potential methane diffusion sites in HKUST-1. As shown in Fig. 3a, HKUST-1 comprises two repeating cyan and orange blocks along with a yellow surface block. In the surface block (Fig. 3b), three types of diffusion sites were identified as follows: surface hydrogen (SH)—a site where H is attached to a surface O created from broken Cu–O bond; surface interconnecting pore (SI)—a site between SH sites; and surface large pore (SL)—a site located between SI sites. For the repeating blocks (Fig. 3c,d), the possible diffusion sites are as follows: metal ion (MI)—a site where metal ion exists; pore near ligand (PL)—a site surrounded by organic ligands; pore near metal ion (PM)—a site adjacent to MI site; and pore near pore (PP)—a site between PM sites. The relationships between these diffusion sites and the OMSs are as follows: In the HKUST-1 structure, the OMSs are the oxygen atoms circled in green (Fig. 3c,d). OMSs are directly present at the MI site. The SI and PM sites are located near the OMSs. The SH, SL, PL, and PP sites are farther from the OMSs. The characteristics of methane diffusion sites are summarized in Table 1.

Using the identified diffusion sites, we analyzed the movement of each methane molecule near the surface and within inactivated and activated HKUST-1. The movement was tracked by monitoring the position of the carbon atom in each methane molecule, recorded every 0.1 ps. First, we investigated the occupancy of the entrance sites through which the methane molecules entered HKUST-1. As shown in Fig. 3b, three diffusion sites are present on the surface blocks: SH, SI, and SL. We counted the number of methane molecules that entered each of these three surface diffusion sites upon their initial entry into HKUST-1, considering cases with 50, 100, 150, or 200 methane molecules. The number of methane molecules entering HKUST-1, as shown in Fig. 2a, follows the same trend across all cases; hence, the occupancies for the four cases were averaged. Figure 4a shows that methane in inactivated and activated HKUST-1 preferentially enters the empty SI and SL sites, rather than the SH site, which is partially blocked by atoms. In activated HKUST-1, more methane molecules entered the SI sites near the OMSs.

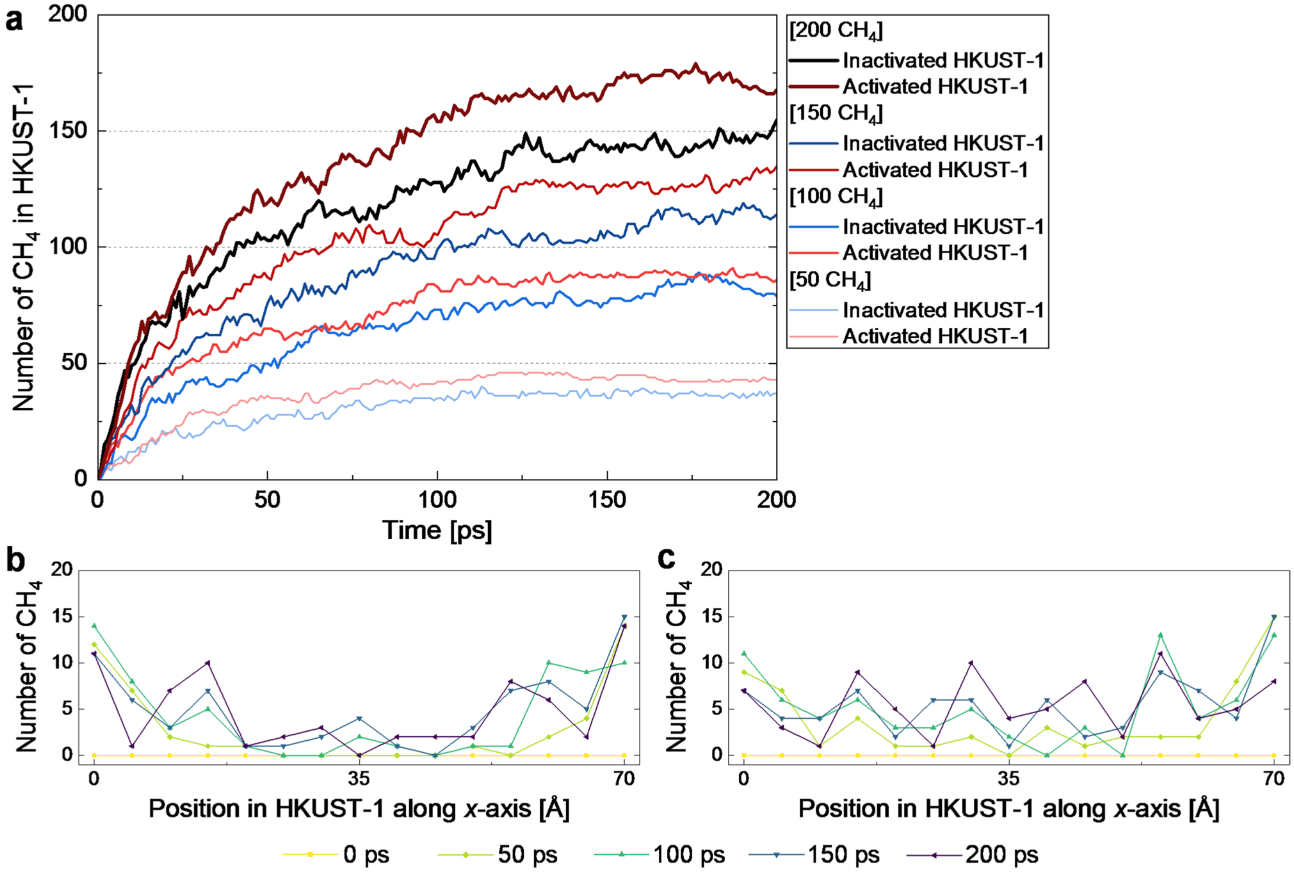

Fig. 2. (**a**) Number of methane molecules entering inactivated and activated HKUST-1 over 200 ps. The total number of methane molecules is 50, 100, 150, or 200. Distribution of methane molecules along the thickness direction of (**b**) inactivated and (**c**) activated HKUST-1 for a system with 100 methane molecules over time. The start and end of each graph correspond to the (_100) and (100) surfaces of HKUST-1, respectively. The center of the graph represents the middle of HKUST-1.

Next, we identified the paths along which the methane molecules moved after entering the surface of HKUST-1. These paths were tracked from the entrance sites to the subsequent diffusion sites, considering cases with 50, 100, 150, or 200 methane molecules. The average occupancy values for the one-time and two-time movement diffusion paths are shown in Fig. 4b,c, respectively. Similar to the entrance sites, methane molecules in both HKUST-1 tended to move along paths consisting of empty sites, specifically from SL to PM (the SL–PM path). In activated HKUST-1, a greater number of methane molecules follow the SI–MI path, which is near and includes OMSs. The two-time movement diffusion path shown in Fig. 4c demonstrates that methane continues to diffuse in the interior of HKUST-1. Similar to the one-time movement diffusion path, the methane molecules in the activated HKUST-1 prefer to move along paths that are near or include OMSs. However, the SI–PM–SI and SL–PM–SI paths show that methane first enters the SI or SL site, moves to the PM site, and then returns to the SI site, indicating that some methane molecules entering the HKUST-1 flow back out. The backflow phenomenon was more pronounced in inactivated HKUST-1 than in activated HKUST-1 (Fig. S1). This backflow likely contributed to the lower methane adsorption observed in the MD simulations for inactivated HKUST-1 (Fig. 2a).

In addition to surface diffusion, we examined the diffusion of methane within the inactivated and activated HKUST-1. Initially, we investigated the distribution of methane at the adsorption sites within inactivated and activated HKUST-1. In Fig. 3c,d, four internal diffusion sites were identified within HKUST-1: MI, PL, PM, and PP. The number of methane molecules at these four sites was counted and summed at intervals of 0.1 ps over 200 ps for cases containing 50, 100, 150, or 200 methane molecules. The average occupancy values for these cases, as shown in Fig. 4d, indicate that methane preferentially occupied the PL site, which is surrounded by organic ligands. By contrast, the MI sites, which are partly blocked by atoms, and the PP sites, where the surrounding area is empty and methane molecules can move easily, exhibited lower methane occupancy. At the PM sites, where four sides are blocked by MI but two sides are connected to the empty PP sites, the methane occupancy was lower than that at the PL sites but higher than that at the MI and PP sites. In activated HKUST-1, the occupancy of methane at the site near (PM) or including OMSs (MI) was higher. The absence of methane at the PP site in activated HKUST-1 was also observed in the experiments of Getzschmann et al.[7]. During these experiments, the methane distribution was also observed at other locations. If these locations were mapped onto

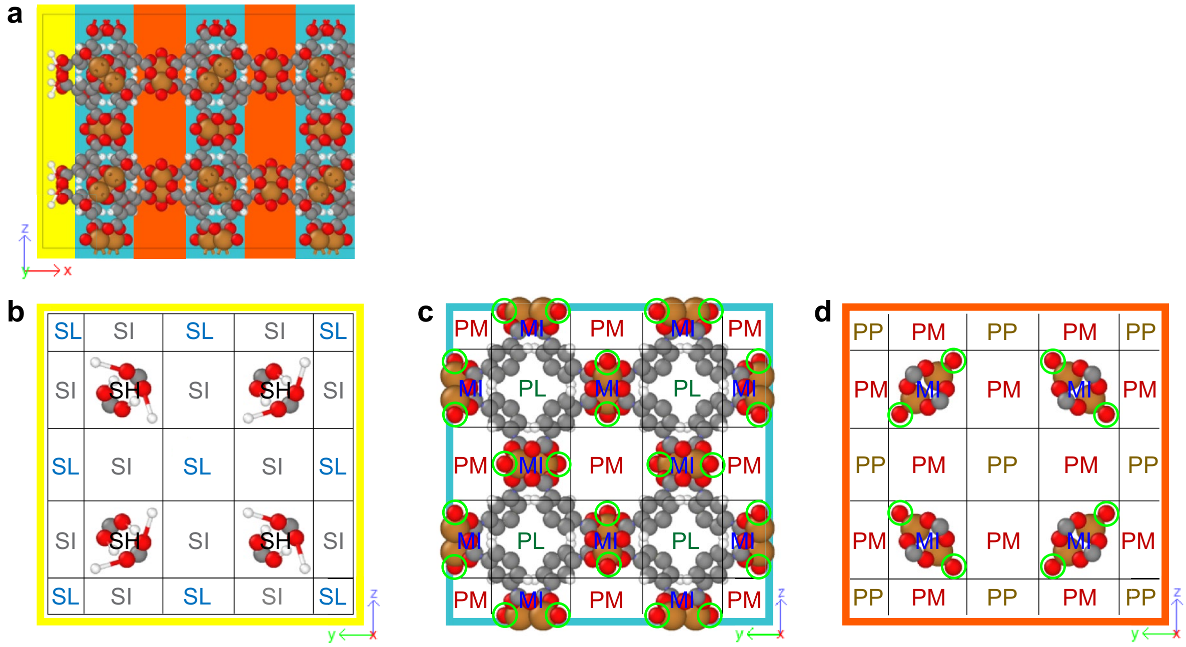

Fig. 3. (**a**) Crystal structure of inactivated HKUST-1 with surfaces on the (_100) plane and divided into yellow, cyan, and orange blocks that are projected onto the (001) plane. Crystal structures of the (**b**) yellow, (**c**) cyan, and (**d**) orange blocks that are projected onto the (100) plane. Colors represent different atoms (orange: Cu, red: O, gray: C, white: H, blue: C in methane, and light blue: H in methane). The oxygen atoms circled in green represent the OMSs in the activated HKUST-1.

Methane diffusion site		Location	Relation to OMS
Surface hydrogen	SH	A site where H is attached to a surface O	Far
Surface interconnecting pore	SI	A site between SH sites	Near
Surface large pore	SL	A site located between SI sites	Far
Metal ion	MI	A site where metal ion exists	Including
Pore near ligand	PL	A site surrounded by organic ligands	Near
Pore near ligand	PM	A site adjacent to MI site	Near
Pore near pore	PP	A site between PM sites	Far

Table 1. List of methane diffusion sites with their abbreviations, locations, and relation to OMS.

the diffusion sites identified in this study, methane occupancy was highest at the PM sites, followed by the PL and MI sites, with the lowest occupancy observed at the PP sites. This trend differs from the results shown in Fig. 4d. The difference would be attributed to the variation in methane loading. In our MD simulations, the methane loading was 1.85 to 7.4 molecules per unit cell, whereas the experiments of Getzschmann et al.[7] involved much higher loadings of 104 and 176 molecules per unit cell. According to GCMC simulations conducted by Chen et al.[8] for activated HKUST-1 with methane loadings of 8, 88, and 176 molecules per unit cell, the adsorption sites are filled by methane molecules in the sequence of PL, PM, MI, and PP. Here, each unit cell of HKUST-1 contains 8 PL sites, 24 PM sites, 24 MI sites, and 8 PP sites. Therefore, in the experiments of Getzschmann et al.[7], the high methane loading results in full saturation of PL and PM sites, with MI sites partially occupied, leading to an overall occupancy trend of PM > PL > MI > PP when considering the number of available sites. In contrast, our study, which had low methane loads, showed that PL sites were saturated, PM sites were partially filled, and MI and PP sites remained largely unoccupied, yielding a different occupancy sequence: PL > PM > MI ≈ PP.

Next, we examined the diffusion paths of the methane molecules moving inside HKUST-1. To minimize the influence of surface interactions, we tracked the diffusion paths from the point where a methane molecule had undergone three-time movement from the entrance sites. This analysis was conducted for cases containing 50, 100, 150, or 200 methane molecules. Because the number of times each methane molecule moved between the diffusion sites varied, we averaged the path counts until at least half of the methane molecules had completed their movements. The average occupancy values of the one-time and two-time movement diffusion paths are shown in Fig. 4e,f, respectively. Notably, for the one-time movement diffusion path shown in Fig. 4e, the occurrence of diffusion into and out of PL (the MI–PL and PL–MI paths) was relatively low compared to other diffusions. The PL site was the primary adsorption site, as shown in Fig. 4d. These results indicate that the high methane

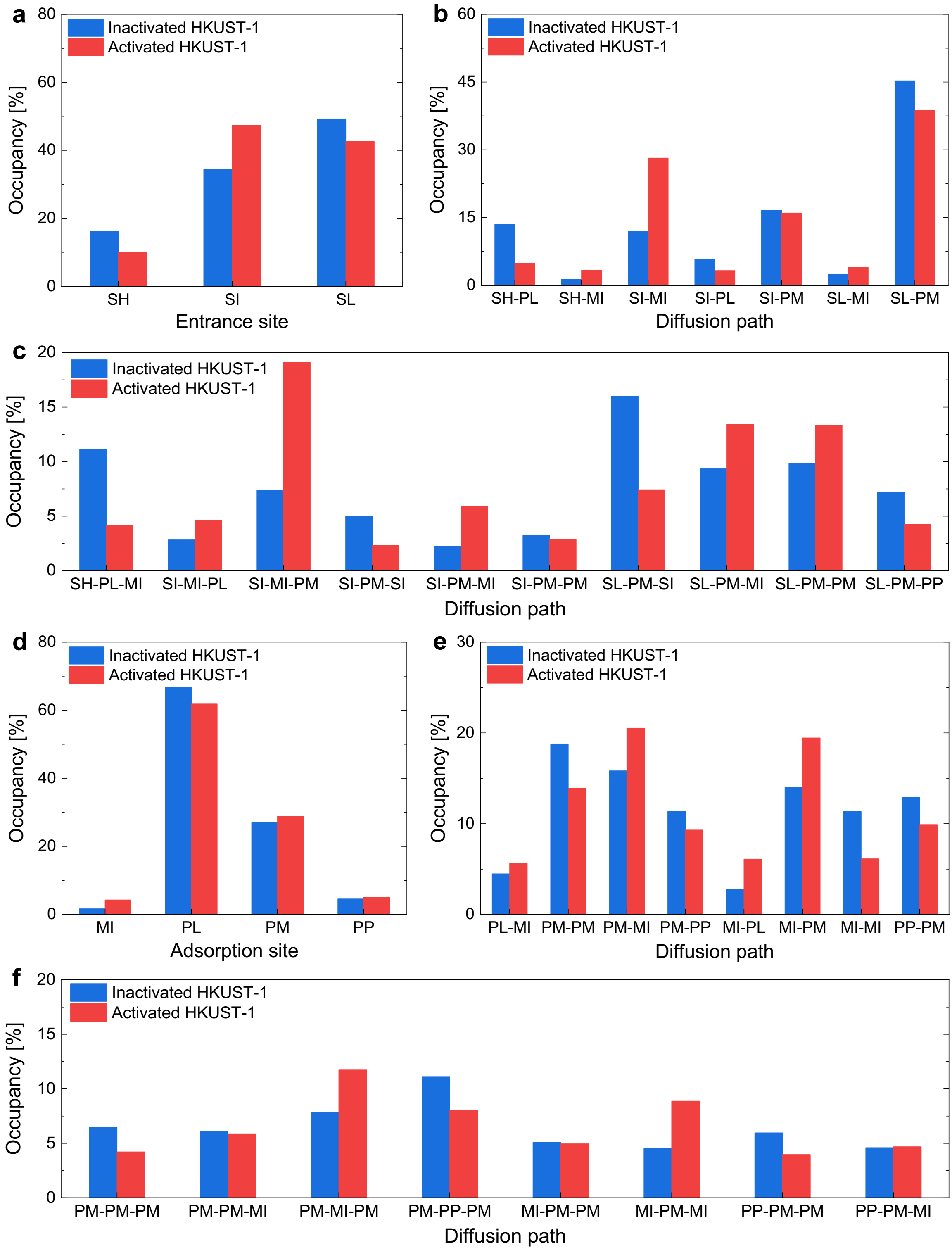

Fig. 4. Occupancy values of (**a**) entrance sites for methane into HKUST-1, (**b**) one-time and (**c**) two-time movement diffusion paths of methane from the entrance sites into HKUST-1, respectively, (**d**) adsorption sites for methane in HKUST-1, and (**e**) one-time and (**f**) two-time movement diffusion paths of methane in HKUST-1, respectively. The values represent the average of the results for cases containing 50, 100, 150, or 200 methane molecules.

occupancy at the PL sites is probably due to restricted diffusion, which prevents the escape of methane once it enters the PL site. This trapping effect is further illustrated in Figs. S3 and S4 in which the methane molecules at the PL sites continued to move but struggled to exit owing to the steric hindrance of the surrounding ligand atoms. Previous studies have also reported the strong methane adsorption capacity of the PL sites. Chen et al.[8] reported that the PL sites provide the most favorable adsorption environment for methane due to enhanced dispersion interactions arising from the small pore size. Similarly, Wu et al.[10] noted that the dimensions and symmetry of the PL cage windows are well-matched with the size of a methane molecule, thereby reinforcing van der Waals interactions and strongly adsorbing methane. These findings, together with the previous studies, confirm that the PL site serves as a highly favorable adsorption site for methane due to the surrounding ligand atoms yielding steric hindrance and van der Waals interactions to methane.

The other one-time movement diffusion mainly occur between the PM site and its adjacent sites, PM, MI, and PP (Fig. 4e). The two-time movement diffusion paths in Fig. 4f also reveal that methane predominantly migrates through the PM site. Notably, the PM site is the second most frequent site where methane is observed, following the PL site (Fig. 4d). However, unlike the PL site, where methane is trapped by surrounding ligands, the PM site functions as the main pathway for methane migration and therefore exhibits a high occupancy rate. This behavior is commonly observed in both HKUST-1. However, there is a difference in preference for the diffusion pathway depending on the presence of OMS. In the case of activated HKUST-1 with OMS, methane migration from PM to PM mainly occurs through the MI site, but when OMS is not present, such movement occurs more frequently through PP sites. This difference is likely driven by the presence of OMSs within the MI sites.

Methane adsorption at both the PL and PM sites is consistently observed in HKUST-1; however, adsorption at the PL site becomes more dominant in inactivated HKUST-1 due to the absence of open metal sites (OMS), which suppresses adsorption at the PM and MI sites. Here, considering the definition of adsorption, which refers to molecules being held in place and unable to move, true methane adsorption can be seen to occur mainly at the PL sites that trap methane, rather than at the PM sites that act as diffusion hubs.

Methane adsorption energy and distance in HKUST-1

The diffusion path analysis raises three questions: (1) why is methane preferentially observed at sites near or including OMSs in activated HKUST-1?, (2) why does methane backflow occur predominantly in inactivated HKUST-1?, and (3) why is the methane diffusion rate of activated HKUST-1 faster than that of inactivated HKUST-1 (Fig. 1) despite of its strong methane adsorption?

To address these questions, we performed first-principles calculations to determine the methane adsorption energy and distance in inactivated and activated HKUST-1. The methane adsorption energy was defined as the energy difference before and after placing one methane molecule in its most stable position within the HKUST-1 sample. The adsorption distance was defined as the distance between the methane and nearest (or second-nearest) the Cu atom in HKUST-1. The calculated adsorption energy and distance are listed in Table 2 and the specific adsorption sites of the methane molecules in inactivated and activated HKUST-1 are shown in Fig. 5.

The first question can be explained by the methane adsorption distances in Table 2. As shown in Fig. 5, the formation of OMS on Cu shifts the methane adsorption site closer to the OMS, resulting in a higher concentration of methane in that region in activated HKUST-1. This shift is consistent with the experiment[7] and is attributed to Coulomb interaction induced by the coordinatively unsaturated OMS. However, despite the Coulomb interaction, the methane adsorption energy in activated HKUST-1 is less negative than in inactivated HKUST-1 without it (Table 2). This counterintuitive result arises because Coulomb interaction scale with the inverse of the distance, while van der Waals interaction scale with the inverse sixth or twelfth power of the distance. The shift in methane adsorption site caused by Coulomb interaction increases the overall intermolecular distance between Cu and methane in activated HKUST-1, thereby reducing its adsorption energy. As a result, despite the strong adsorption tendency near OMSs, activated HKUST-1 exhibits reduced methane adsorption energy, leading to a faster methane diffusion rate. This provides an answer to question (3).

Lastly, regarding question (2), the methane backflow can be attributed to the location of the stable methane adsorption sites. In inactivated HKUST-1, the stable adsorption site is the PM site, which is located in the orange block (Fig. 3d). As previously mentioned, HKUST-1 exhibits a NaCl-type structure with stable {100} surface planes, which correspond to the yellow surface block in Fig. 3b. Given the overall framework structure (Fig. 3a), the PM site in the orange block aligns with the SI site on the surface, making the SI site a favorable adsorption site for methane. As a result, some methane molecules that entered the structure left out through the surface. By contrast, in activated HKUST-1, the stable methane adsorption site lies between the PM site in the cyan block (Fig. 3c) and the PM site in the orange block (Fig. 3d). This spatial shift makes the SI site on the surface (Fig. 3b) less favorable for methane adsorption, allowing more methane molecules to continue diffusing into the interior of the structure rather than escaping.

		Inactivated HKUST-1	Activated HKUST-1 with one OMS
Methane adsorption energy		− 0.28	− 0.15
Methane adsorption distance	1st nearest	3.62	3.02
	2nd nearest	3.92	5.65

Table 2. Methane adsorption energy (eV) and distance (Å) of inactivated HKUST-1 and activated HKUST-1 with an OMS calculated using first-principles calculation.

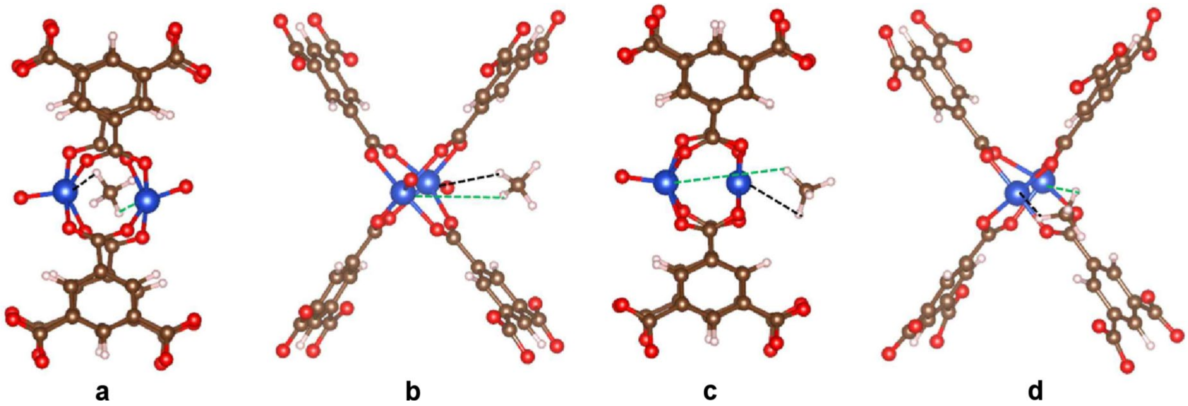

Fig. 5. (**a**,**c**) Back view and (**b**,**d**) dynamic view of (**a**,**b**) $1\times1\times1$ inactivated and (**c**,**d**) activated HKUST-1 with one OMS, each containing a methane molecule after relaxation at 0 K. Colors represent different elements (blue: Cu, red: O, brown: C, and white: H). The dashed black and green lines indicate the first and second nearest distances between methane and Cu, respectively.

Diffusion behaviors of methane in HKUST-1

Our simulation results indicate that in both HKUST-1, methane enters the empty surface sites (SI, SL) and diffused into the frameworks. Once inside, it primarily migrates through the PM site. This is likely because the MI site is too confined to allow easy movement, and the PP site is too open to provide sufficient van der Waals interactions. In contrast, the PM site is located near the MI site, allowing for favorable van der Waals interactions (and, in the case of activated HKUST-1, additional Coulomb interactions due to the presence of OMSs), while still offering enough empty space to facilitate methane migration. During migration, some methane is trapped in the pores surrounded by organic ligands (PL) that cause steric hindrance and van der Waals interactions. This trapping phenomenon may contribute the high methane adsorption of HKUST-1.

In inactivated HKUST-1, stable methane adsorption sites are also present on the {100} surface planes, leading to methane backflow at the surface and thereby reducing the amount of methane entering the interior. Moreover, the methane that does enter diffuses slowly due to its strong adsorption energy, making inactivated HKUST-1 less favorable for methane adsorption.

By contrast, in activated HKUST-1, the stable methane adsorption sites shift toward the OMSs due to Coulomb interactions, resulting in a higher probability of methane being adsorbed at or near these sites. This shift renders the surface sites less favorable for adsorption, allowing more methane to diffuse into the interior. In addition, the relatively weaker adsorption energy facilitates faster diffusion. As a result, activated HKUST-1 exhibits more favorable for methane adsorption.

Experimental measurement of methane uptake of HKUST-1

To verify whether the difference in adsorption behavior also occurs experimentally, we measured the methane uptake for both HKUST-1 under identical temperature and pressure conditions. For the measurement, inactivated and activated HKUST-1 samples with exposed {001} facets were synthesized. The synthesized powders appeared sky-blue for inactivated HKUST-1 and purple for activated HKUST-1 (Fig. 6a,b), consistent with the known color change associated with the degree of activation[19]. Inactivated HKUST-1 particles exhibited a uniform cubic morphology with an average particle size of 3.5 μm (Fig. 6c). After activation of OMSs, the morphology of activated HKUST-1 particles retained their cubic shape (Fig. 6d). Powder X-ray diffraction (PXRD) patterns of both samples closely matched that of simulated HKUST-1 (Fig. 6e). Both PXRD patterns show a significant increase in the intensity of the peak corresponding to the (004) crystal plane, confirming the successful synthesis of cubic-shaped HKUST-1with exposed {001} facets[20]. Notably, the intensity of the peak corresponding to the (440) crystal plane of inactivated HKUST-1 was higher than that of activated HKUST-1, indicating the presence of the solvent molecules. Additionally, the main peak corresponding to the (222) crystal plane of inactivated HKUST-1 was shifted toward lower 2θ angles, meaning an expansion of the crystal lattice owing to the presence of solvent molecules[21] (Fig. S5a). Detailed analysis of the synthesized HKUST-1 are described in the Supplementary Information and Fig. S5.

Methane adsorption isotherms were measured at 300 K after heat treatment under dynamic vacuum at 373 K for 12 h. As shown in Fig. 6f, inactivated HKUST-1 exhibited a methane uptake of 10.66 mg/g, while activated HKUST-1 showed a higher uptake of 12.35 mg/g. This experimentally observed difference in uptake strongly supports the simulation results, indicating that activated HKUST-1 is more favorable for methane adsorption than inactivated HKUST-1.

Future works

In the present study, a simplified inactivated HKUST-1 model, in which only O atoms were coordinated to the Cu sites, was employed in order to focus on the effect of OMSs on the diffusion behavior of methane. However, in actual inactivated HKUST-1, H_2O or *N*, *N*-dimethylformamide (DMF) molecules are coordinated to the Cu sites instead of O atoms[21,22]. Therefore, future work will extend the present approach by explicitly incorporating H_2O-

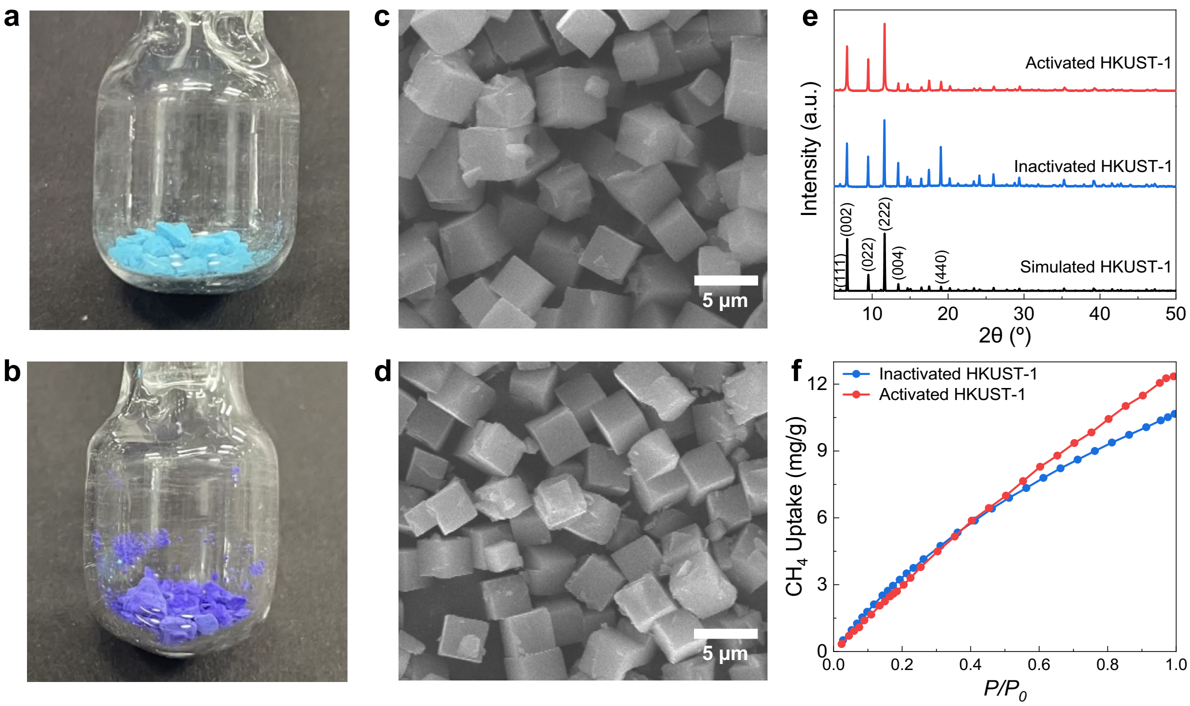

Fig. 6. Synthesized (**a**) inactivated HKUST-1 and (**b**) activated HKUST-1 powders under vacuum conditions. Field-emission scanning electron microscopy (FE-SEM) images of (**c**) inactivated HKUST-1 and (**d**) activated HKUST-1. (**e**) PXRD patterns of the as-synthesized inactivated HKUST-1 and activated HKUST-1. (**f**) CH_4 uptake capacities of inactivated and activated HKUST-1 measured at 300 K.

or DMF-coordinated inactivated HKUST-1 models to investigate their influence on methane diffusion behavior in a more realistic manner. In addition, motivated by the NaCl-type crystal structure of HKUST-1, the present study focused exclusively on samples exposing {100} surfaces. In future studies, surface models exposing other low-index planes, such as {111} and {110}, will be constructed to systematically examine the effect of surface orientation on methane diffusion behavior.

Methods
MD simulations
Performing MD simulations requires samples and force fields. Those samples for HKUST-1 were created using the CIF data reported by Yakovenko et al.[23]. This CIF structure corresponds to the inactivated HKUST-1 (Fig. 7a). The structure of the activated HKUST-1 was generated by removing one O atom facing the hole from the Cu center, exposing an OMS, as shown in Fig. 7b. It should be mentioned here that in actual inactivated HKUST-1, H_2O or DMF molecules are bound to the OMS positions of Cu atoms instead of O atoms[21,22]. However, including coordinated H_2O or DMF molecules in the simulation increases the structural complexity of HKUST-1, which in turn makes the analysis of methane diffusion behavior considerably more difficult. Therefore, we intentionally simplified the inactivated HKUST-1 model by coordinating only O atoms to the Cu sites to focus on the diffusion behavior of methane in HKUST-1 in the absence of OMS. Methane was modeled as an all-atom type containing one C and four H atoms (Fig. 7c) to more accurately capture its adsorption and diffusion behaviors.

Force fields capable of describing the interactions between methane and OMSs in activated HKUST-1 have been developed by Chen et al.[8], and Koh et al.[12], both of which employed a united-atom description of methane. In contrast, as illustrated in Fig. 7c, we adopted an all-atom model for methane in this study. Therefore, the previously developed force fields were not used. Instead, the force field for methane was adopted from OPLS-AA[24] to explicitly account for the H atoms and define the partial charges for the C and H atoms in a methane molecule. For HKUST-1, we used the force field developed by Zhao et al.[25], which reproduces the crystal structure, negative thermal expansion, vibrational properties, and CO_2 adsorption while treating the framework as flexible. The validation of this force field is described in the Supplementary Information and Tables S1. Because this force field was originally designed for activated HKUST-1, modifications were necessary for inactivated HKUST-1 in which an additional O atom (denoted as O3 in Fig. S6) is bonded to Cu. For surface creation, the Cu–O bonds on the (100) and (_100) planes were broken, and new H atoms (denoted as H2) were attached to the broken O atoms. To describe O3 and H2, we introduced the non-bonded, bond, and angle parameters for them. The detailed parameterization process and developed force field parameters are described in the Supplementary Information and Tables S2 and S3, respectively.

The diffusion coefficient of methane for HKUST-1 was calculated using the mean square displacement (MSD) over time. To calculate the MSD, we created 3×3×3 unit cell samples of inactivated and activated HKUST-1 that satisfied the periodic boundary conditions in all directions. Both samples were first relaxed at

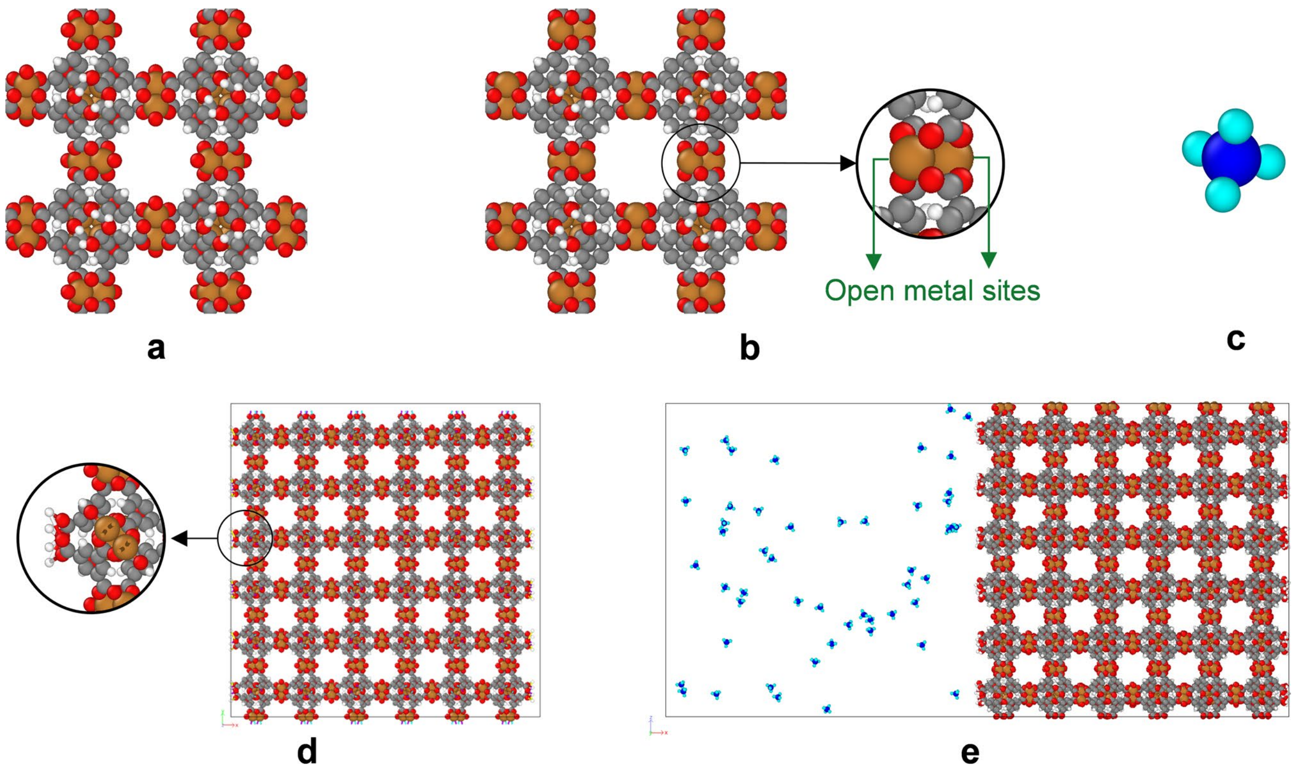

Fig. 7. Crystal structures of (**a**) inactivated HKUST-1, (**b**) activated HKUST-1, and (**c**) a methane molecule. (**d**) A 3 × 3 × 3 unit cell of inactivated HKUST-1 with exposed surfaces on (100) and (_100) planes projected onto the (001) plane. (**e**) A snapshot of the simulation of the 3 × 3 × 3 inactivated HKUST-1 framework with 50 methane molecules at 300 K. Colors indicate different elements (orange: Cu, red: O, gray: C, white: H, blue: C in methane, and light blue: H in methane).

0 K using the conjugate gradient method. Then, the NPT ensemble was applied at 300 K and one atm for 30 ps to allow the system to reach its equilibrium volume. 135 methane molecules (5 methane molecules per unit cell) were randomly introduced into each HKUST-1 sample, and the NVT ensemble was applied at 300 K for 30 ps to achieve equilibrium pressure. Finally, the NVT ensemble was applied for 20 ns to calculate the MSD, thereby avoiding possible energy drift associated with the use of the NVE ensemble. The MSD was calculated in three dimensions using the production trajectory after discarding the initial 30 ps used for equilibration. To enhance the statistical reliability of the MSD calculation, three independent MSD calculations were performed under identical conditions but with different initial configurations, and the results were averaged to calculate the diffusion coefficient.

The diffusion path of methane in HKUST-1 was investigated using inactivated and activated HKUST-1 samples with a 3 × 3 × 3 unit cell size with surfaces on the (100) and (_100) planes. These surface orientations were selected based on the crystal structure of HKUST-1, which belongs to the *Fm_3m* space group, corresponding to the NaCl-type structure. For the NaCl-type structures, the {100} surface generally exhibits lower surface energy than other low-index surfaces such as {111} and {110} surfaces, leading to the preferential exposure of {100} facets and a cubic morphology. Accordingly, the (100) and (_100) planes along the *x*-axis were chosen as the surface planes for the present study. This choice is further supported by our experimental observations, in which the synthesized HKUST-1 crystals predominantly exhibit a cubic morphology (Fig. 6c,d). To create a surface, the bonds intersecting the (100) and (_100) planes should be broken. In creating a stable surface, it is important to minimize the surface energy, which could be expressed as being proportional to the number of broken bonds and their corresponding bond energies. In the present case, although the Cu–O bond has a relatively high bond energy compared to other bonds in HKUST-1 (Table S2), the number of Cu–O bonds intersecting the selected planes is smaller than that of alternative bonds. Based on this consideration, we generated the (100) and (_100) surfaces by breaking Cu–O bonds, leaving an O-terminated layer on the surface. The O atoms whose bonds to Cu were broken cannot remain undercoordinated; therefore, H atoms were attached to the broken Cu–O bond sites to chemically saturate the surface (Fig. 7d). The other two directions were set to satisfy the periodic boundary conditions. As shown in the procedure in Fig. S7, these samples were first relaxed at 0 K using the conjugate gradient method, and then equilibrated using the NPT ensemble at 300 K and 1 atm for 30 ps to obtain the equilibrium volume under these conditions. Next, an empty space with a thickness equal to the original sample dimension was added on one side of the slab normal to the surface. Methane molecules (50, 100, 150, or 200 molecules) were then randomly inserted into the empty space to make methane molecules diffuse toward the HKUST-1 slab. The NVT ensemble was then applied at 300 K for 200 ps to observe the diffusion path of methane in both HKUST-1. The occupancy of the diffusion sites was determined by analyzing the positions of the methane molecules at 0.1 ps intervals, with the results averaged across these intervals. The location of the methane molecules was based on the position of the C atom. All MD simulations were performed using the large atomic/molecular massively parallel simulator (LAMMPS) software[26] and visualized with the OVITO software[27]. The time step was set to one fs. Periodic boundary conditions applied in all directions. The present NPT and NVT simulations employed a Nosé–Hoover style thermostat and barostat. The target temperature and pressure were set to 300 K and 1 atm, respectively. The thermostat coupling time constant (τT) is 10 fs and the barostat coupling time constant (τP) is 100 fs. The Lennard-Jones interactions employed a cutoff radius of 12 Å, and their parameters were mixed according to the geometric mixing rule. Long-range electrostatic interactions were handled using the Ewald summation technique, applying a desired relative error of 0.0001 for the force. Those MD simulation parameters are summarized in Table S4.

First-principles calculations

The methane adsorption energy and distance for inactivated and activated HKUST-1 were calculated using first-principles calculations with the Vienna ab initio simulation package (VASP)[28,29] and the projector augmented wave (PAW) method[30]. The generalized-gradient approximation (GGA) with the Perdew–Burke–Ernzerhof (PBE) functional was used for the exchange-correlation functional[31]. A plane-wave basis set with a cut-off energy of 500 eV was employed, and the Brillouin zone was sampled at the gamma center, with a $1 \times 1 \times 1$ mesh. The energy and maximal force convergence criteria were set to 10^{-6} eV and 0.02 eV/Å, respectively. The van der Waals dispersion correction was applied using the DFT-D2 method[32,33]. The inactivated HKUST-1 sample used for the calculation comprised a $1 \times 1 \times 1$ unit cell with 672 atoms, corresponding to the crystal structure shown in Fig. 7a. The activated HKUST-1 sample was created by removing one O atom facing the hole from the Cu center in the $1 \times 1 \times 1$ inactivated HKUST-1 sample to create an OMS. Similar to the MD simulation, the methane sample comprised all-atom methane with one C and four H atoms.

Experiments

Copper(Ⅱ) nitrate trihydrate ($Cu(NO_3)_2{\cdot}3H_2O$, puriss. p.a., 99%–103%) and BTC (95%) were purchased from Sigma-Aldrich (USA). Other solvents such as DMF (99.5%), methanol (99.5%), ethanol (99.9%), acetonitrile (99.5%), dichloromethane (99.8%), and pyridine (99.5%) were purchased from Samchun Chemical Co., Ltd. (South Korea). All solvents and chemicals were used as received unless otherwise specified.

The synthesis of inactivated HKUST-1 was carried out following a modified procedure from the literature[34]. Solutions were prepared by dissolving $Cu(NO_3)_2{\cdot}3H_2O$ (0.4 g, 1.66 mmol) and BTC (0.145 g, 0.69 mmol) in 100 mL of DMF. Separately, a pyridine solution was prepared by dissolving pyridine (1.42 mL, 17.5 mmol) in 50 mL of deionized water. The ligand and pyridine solutions were combined with 25 mL of ethanol, after which the metal solution was added to the mixture. The resulting mixture was then placed in an oil bath at 303 K for 30 min. The precipitated crystals of inactivated HKUST-1 were isolated and washed several times with DMF. Activated HKUST-1 was obtained by further washing the as-synthesized inactivated HKUST-1 crystals with acetonitrile and dichloromethane to activate OMSs that were occupied by coordination bonds, following a reported procedure[35].

The morphology of the synthesized inactivated and activated HKUST-1 was examined using field-emission scanning electron microscopy (FE-SEM, JSM 7900 F, JEOL Ltd., Japan). The crystal structures of the synthesized HKUST-1 were characterized by powder X-ray diffraction (PXRD, SmartLab XE diffractometer, Rigaku, Japan) with a Cu–Kα radiation source (λ = 1.5406 Å) operated at 45 kV and 200 mA. Attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopy (Nicolet iS10 FTIR spectrometer, Thermo Fisher Scientific Inc., USA) was used to confirm the successful synthesis of both HKUST-1. N_2 adsorption–desorption isotherms at 77 K and CH_4 adsorption isotherms at 300 K were obtained using gas adsorption analyzers (BELSORP MAX X and BELSORP MINI, MicrotracBEL, Japan). The data were analyzed using the Brunauer–Emmett–Teller (BET) method. All adsorption isotherms were measured after pretreatment under a dynamic vacuum at 373 K for 12 h. Thermogravimetric analysis (TGA, TGA 55, TA Instruments, USA) was performed to assess the thermal stability of HKUST-1 and the presence of guest or solvent molecules in its pores.

Data availability

The data generated and analyzed during the current study are available from the corresponding author on reasonable request.

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References

1. Allen, D. T. Methane emissions from natural gas production and use: Reconciling bottom-up and top-down measurements. *Curr. Opin. Chem. Eng.* **5**, 78–83 (2014).
2. Mahmoud, E. Evolution of the design of CH4 adsorbents. *Surfaces* **3**, 433–466 (2020).
3. Gutiérrez-Serpa, A., Pacheco-Fernández, I., Pasán, J. & Pino, V. Metal–organic frameworks as key materials for solid-phase microextraction devices—A review. *Separations.* **6**, 1–29 (2019).
4. Uzun, A. & Keskin, S. Site characteristics in metal organic frameworks for gas adsorption. *Prog. Surf. Sci.* **89**, 56–79 (2014).
5. Chong, K. C. et al. A review of HKUST-1 metal-organic frameworks in gas adsorption. *IOP Conf. Ser. Earth Environ. Sci* **1135**, (2023).
6. Lin, Y., Kong, C., Zhang, Q. & Chen, L. Metal-organic frameworks for carbon dioxide capture and methane storage. *Adv. Energy Mater.* **7**, (2017).

7. Getzschmann, J. et al. Methane storage mechanism in the metal-organic framework Cu 3(btc)2: An in situ neutron diffraction study. *Microporous Mesoporous Mater.* **136**, 50–58 (2010).
8. Chen, L., Grajciar, L., Nachtigall, P. & Düren, T. Accurate prediction of methane adsorption in a metal-organic framework with unsaturated metal sites by direct implementation of an ab initio derived potential energy surface in GCMC simulation. *J. Phys. Chem. C.* **115**, 23074–23080 (2011).
9. Zhang, Z., Zhao, Y., Gong, Q., Lib, Z. & Li, J. MOFs for CO2 capture and separation from flue gas mixtures: The effect of multifunctional sites on their adsorption capacity and selectivity. *Chem. Commun.* **49**, 653–661 (2013).
10. Wu, H. et al. Metal-organic frameworks with exceptionally high methane uptake: Where and how is methane stored? *Chem. - Eur. J.* **16**, 5205–5214 (2010).
11. Rappé, A. K., Casewit, C. J., Colwell, K. S., Goddard, W. A. & Skiff, W. M. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *J. Am. Chem. Soc.* **114**, 10024–10035 (1992).
12. Koh, H. S., Rana, M. K., Wong-Foy, A. G. & Siegel, D. J. Predicting methane storage in open-metal-site metal-organic frameworks. *J. Phys. Chem. C.* **119**, 13451–13458 (2015).
13. Wang, J. & Hou, T. Application of molecular dynamics simulations in molecular property prediction II: Diffusion coefficient. *J. Comput. Chem.* **32**, 3505–3519 (2011).
14. Kondrat, S. Physics and modelling of intracellular diffusion (2018).
15. Song, Y., Jiang, B. & Li, F. L. Molecular dynamic simulations of selective self-diffusion of CH4/CO2/H2O/N2 in coal. *IOP Conf. Ser. Mater. Sci. Eng.* **213**, (2017).
16. Babaei, H., McGaughey, A. J. H. & Wilmer, C. E. Transient mass and thermal transport during methane adsorption into the metal-organic framework HKUST-1. *ACS Appl. Mater. Interfaces.* **10**, 2400–2406 (2018).
17. Wang, Z. et al. Molecular insights into hybrid CH4 physisorption-hydrate growth in hydrophobic metal–organic framework ZIF-8: Implications for CH4 storage. *Chem. Eng. J.* **430**, 132901 (2022).
18. Awadallah-F, A., Hillman, F., Al-Muhtaseb, S. A. & Jeong, H. K. Adsorption equilibrium and kinetics of nitrogen, methane and carbon dioxide gases onto ZIF-8, Cu10%/ZIF-8, and Cu30%/ZIF-8. *Ind. Eng. Chem. Res.* **58**, 6653–6661 (2019).
19. Ahmed, H. et al. Acoustomicrofluidic assembly of oriented and simultaneously activated metal–organic frameworks. *Nat. Commun.* **10**, 1–10 (2019).
20. Mao, Y. et al. Specific oriented metal-organic framework membranes and their facet-tuned separation performance. *ACS Appl. Mater. Interfaces.* **6**, 15676–15685 (2014).
21. Alfonso-Herrera, L. A. et al. Elucidating structural stability, bandgap, and photocatalytic hydrogen evolution of (H2O/DMF)@ HKUST-1 host-guest systems. *Chempluschem.* **89**, (2024).
22. Prestipino, C. et al. Local structure of framework Cu(II) in HKUST-1 metallorganic framework: Spectroscopic characterization upon activation and interaction with adsorbates. *Chem. Mater.* **18**, 1337–1346 (2006).
23. Yakovenko, A. A., Reibenspies, J. H., Bhuvanesh, N. & Zhou, H. C. Generation and applications of structure envelopes for porous metal-organic frameworks. *J. Appl. Crystallogr.* **46**, 346–353 (2013).
24. Jorgensen, W. L., Maxwell, D. S. & Tirado-Rives, J. Development and testing of the OPLS all-atom force field on conformational energetics and properties of organic liquids. *J. Am. Chem. Soc.* **118**, 11225–11236 (1996).
25. Zhao, L. et al. A force field for dynamic Cu-BTC metal-organic framework. *J. Mol. Model.* **17**, 227–234 (2011).
26. Plimpton, S. Fast parallel algorithms for short-range molecular dynamics. *J. Comput. Phys.* **117**, 1–19 (1995).
27. Stukowski, A. Visualization and analysis of atomistic simulation data with OVITO-the Open Visualization Tool. *Model Simul. Mater. Sci. Eng.* **18**, (2010).
28. Kresse, G. & Hafner, J. Ab initio molecular dynamics for liquid metals. *Phys. Rev. B.* **47**, 558–561 (1993).
29. Kresse, G. & Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B Condens. Matter Mater. Phys.* **54**, 11169–11186 (1996).
30. Blöchl, P. E. Projector augmented-wave method. *Phys. Rev. B Condens. Matter Mater. Phys.* **50**, 17953 (1994).
31. Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
32. Stefan Grimme. Semiempirical GGA-type density functional constructed with a long-range dispersion correction. *J. Comput. Chem.* **27**, 1787–1799 (2006).
33. Koh, H. S., Rana, M. K., Hwang, J. & Siegel, D. J. Thermodynamic screening of metal-substituted MOFs for carbon capture. *Phys. Chem. Chem. Phys.* **15**, 4573–4581 (2013).
34. Lee, S., Lee, G. & Oh, M. Lattice-guided construction and harvest of a naturally nonpreferred metal-organic framework. *ACS Nano.* **15**, 17907–17916 (2021).
35. Bae, J. et al. Multiple coordination exchanges for room-temperature activation of open-metal sites in metal-organic frameworks. *ACS Appl. Mater. Interfaces.* **9**, 24743–24752 (2017).

Author contributions

H.-S.J. wrote the main manuscript and conducted MD simulations. E.C. and H.J.L. conducted experiments and wrote experimental parts in manuscript. G.K. and B.-H.K. conducted first-principles calculations. All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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